

Algebra II with Trigonometry

Chapter 5.5

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:

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1. The Ambiguous Case Which specific oblique triangle case (using abbreviations like ASA, SAS, SSA, SSS) is known as the "ambiguous case" where zero, one, or two triangles might exist?

2. Core Principle According to the Law of Sines definition, the length of any side of a triangle is proportional to the sine of which specific angle?

3. Calculating Triangle Height In an ambiguous case triangle where angle A is 30° and the adjacent side b is 12, what is the height (h) of the triangle? (Recall that $\sin 30^\circ = 0.5$)

4. Obtuse Angle Evaluation (Part 1) You are given triangle dimensions where angle A is an obtuse 120° , side $a = 15$, and side $b = 20$. Based on the rules for obtuse angles, how many possible triangles can be formed?

5. Obtuse Angle Evaluation (Part 2) You are given triangle dimensions where angle A is an obtuse 110° , side $a = 28$, and side $b = 15$. How many possible triangles can be formed?

6. Acute Angle Limits (Zero Solutions) You are evaluating an acute triangle and calculate the height (h) to be 8. If the side opposite your given angle is $a = 5$, how many triangles can be formed?

7. Acute Angle Limits (Two Solutions) In an acute triangle, you calculate the height (h) to be 6. If the given side $b = 10$, and the side opposite the given angle is $a = 8$, how many possible triangles are there?

8. Constant Ratios In triangle ABC, if side $a = 20$ and $\sin A = 0.5$, what must the value of the ratio $b / \sin B$ equal?

9. The Supplement Check When checking for a second possible triangle in the ambiguous case, you must verify that the sum of the original angle and the new supplementary angle does not exceed what degree measure?

10. Maximum Sine Value While solving for an unknown angle in an ambiguous case, you calculate that the sine of the angle equals 1.5. Without doing any further work, how many valid triangles exist?